

between this and the fact that the development of the underlying wave function for the whole universe is unique.

Summarizing:

Definition 1. A *universe* U is a chain of states (one state U_t for each moment of time t), with the property that the transition between adjacent states is always possible.

Definition 2. A *multiverse* M is the set of all possible universes U in the sense of Definition 1 together with a probability measure on this set.

It may of course be said that quantum mechanics should allow for transitions between all kinds of states, although the probability for most such transitions may be extremely small. In this extremely simplified treatment, I will assume that for a given state at a given moment of time t , the dynamical laws will only permit transitions to a very limited number of states at the previous and next moments, which will make the probabilistic part of the investigation particularly simple. However, modifications are called for near the endpoints (the Big Bang and the Big Crunch); see Section 5.

As it stands, the model presented so far is too simple to generate any results. In fact, there are no observable differences at all between the states, which mean that there are no measurable variables which could be related to the (so far non-specified) dynamics.

There are of course many different variables which we can choose to enrich this structure, and which ones to choose must depend on what properties we want to explain. For explaining the second law of thermodynamics, the obvious choice is the entropy.

4. Entropy

According to Boltzmann, the total entropy of a certain macro-state at a certain time is given by

$$S = k_B \ln \Omega, \quad (2)$$

or inversely

$$\Omega = W^S, \quad \text{with} \quad W = e^{1/k_B}, \quad (3)$$

where Ω denotes the number of corresponding micro-states and k_B is Boltzmann's constant.

This formula was from the beginning derived for simple cases, like an ideal gas. Nevertheless, it does represent a kind of universal truth in statistical mechanics: the number of possible micro-states corresponding to a given macro-state grows exponentially with the entropy. Although there are many complications when one tries to consider the entropy of the universe as a whole, I will still take it as the starting point for the discussion that the entropy (at a given time t) is an exponential function of the total entropy as in (3). A more difficult question is if and how the constant W may vary with time, but for the purpose of the present paper, I will simply let it be constant.

One may of course argue that this can only be true when the universe is still quite ordered and the entropy is very far from reaching its maximum. But this is certainly what the situation is like in our universe today, and according to the computations in [10, 11], it would take an almost incredibly long time to reach such a state of maximal entropy. Thus, it will in the following be taken for granted that this time is much longer than the life-span of our universe.